

Alzheimer Disease



Prepared by:

Dr. Mayada Taha Mahmmoud

Assistant Professor of community health nursing

F.N. Helwan University

Out line

- Introduction
- Definition
- Risk factors
- The different between normal and Alzheimer brain
- Etiology
- Stages
- Diagnosis
- Treatment
- Complications
- Nursing role

Introduction

AD is common with very old age, only about 3% of elders age 65 to 75 years have AD., about 20% of those age 75 to 84 years and nearly 50% of those age 85 years and older have AD. The average length of illness from the onset of symptoms until death is approximately 8 to 10 years (some have lived up to 2- years).

Definition

It is a chronic progressive degenerative disorder of cerebral cortex (especially frontal lobe) causes gradual loss of memory, with loss of at least one other cognitive function such as language, difficulty with abstract thinking, judgment or orientation.

Risk factors

- Increase age
- Family history

- Female gender
- Presence of down syndrome
- Low formal education

Etiology

The exact cause of AD is unknown, several theories have been proposed and included the following:

1. ***Genetic theory***: AD be genetically transmitted in some cases, mutations in the gene chromosome 21 have been associated with development of the disease
2. ***Slow virus theory***: slow growing virus that attacks brain cells, this virus changes the brain chemistry and causes brain cell changes that result in AD.
3. ***Aluminum theory***: exposure to heavy metal such as Aluminum, lead, ...etc.
4. ***Cholinergic theory***: this theory suggests that decrease of acetylcholine occurs in the brain of patients with AD, causing memory loss and other cognitive deficits associated with AD.
5. ***Amyloid theory***: Beta- amyloid protein deposits, known as plaques, are found in the brain of those diagnosed with AD.
6. ***Autoimmune theory***: abnormality in the immune system can lead to the accumulation of damage to the brain.

Stages

Stage 1: early stage, forgetfulness, uncertainly phase, (2-5y)

- In this stage the individual have difficulty in remembering recent, they forget where object are placed (key, eye glasses) and have difficulty remembering names than in the past.
- Forgetting events such as birthday, names, long pauses between words and sentences

- Change in personality
- Change in emotions e.g.: depression

“In this stage the patients able to work normally”

Stage 2: Middle stage, confutional phase, (2-8y)

In this stage, the client complain from:

- Delusion and hallucination
- Wandering
- Resistance to help
- The patient may have difficulty understanding, concentration and making decision

Gait changes, a person will being to shuffle.

Stage 3: dementia phase, (1-3y)

- In this stage the elderly person can't able to remembering the family members
- Psychosis symptoms as hallucination and delusion were occur

Stage 4: later stage

- In this stage the patient loss the ability to function mentally and physically complete memory loss, doesn't recognize family members
- Difficult swallowing
- Incontinence of bowel and bladder
- The patient become completely dependent on others for care

Diagnosis

- Complete health history

- Physical examination
- Neurologic and mental status assessment
- Other tests include blood and urine analysis
- Electrogram, MRI and computed tomography (CT)

The American psychiatric association referred that the diagnosis of Alzheimer include:

- 1) Memory impairment
- 2) One of the following cognitive disturbance:
 - ✓ Aphasia: forgetting simple words and can't substituting one another with a similar word
 - ✓ Apraxia: difficulty in performing motor activity despite intact physical functioning
 - ✓ Agnosia: failure to recognize objects in spite of intact sensory functioning

Treatment

All treatment in AD to delay the symptoms appear, the treatment as the following:

- Tacrin hydrochloride and aricept
- Vitamin E 1000 IU twice/ day
- Other drugs such as anti-psychotic and antidepressants

Complications

- Pneumonia and other infections
- Malnutrition and dehydration
- Aspiration

Nursing role

☐ Reality orientation: orient the client by (person, time and place)

☐ Nutrition:

- Encourage sufficient fluids intake and adequate nutrition
- Provide balanced diet with adequate fibers
- Avoid stimulants such as coffee, tea, cola and chocolate
- Give the patient semisolid foods if he has dysphagia
- Plan a high caloric diet if the patient is hyperactive

☐ Environment:

Provide a safe environment , avoid sudden change in the environment, provide good lightening and limiting unnecessary noise

☐ Hygiene: maintain patient's personal hygiene

☐ Elimination:

- Interesting to avoid constipation,
- Limit the fluids at bed time to prevent night incontinence

☐ Medication:

- the nurse should be provide the treatment at the right time, right dose, right rout, right patient and right medication

- Communication: use soft tone, simple sentences and calm manner when speaking
- Maintain eye to eye contact
- Allow sufficient time to answer questions

❑ Activity:

- Because the patient has a short attention span, provide task oriented activities based on things the patient is familiar with
- Break tasks, into parts.

❑ Exercises

❑ Coping with behavior problems:

- Wandering
- Sundown syndrome
- Resisting care